

[Irrigation Management](#)

Water Management

For the first two years from planting, irrigate at the rate of 45 litres of water/seedling, once in 4 days during dry summer months. *(Click the respective state to navigate this page)*

- [Tamil Nadu](#)
- [Kerala](#)
- [Karnataka](#)

Tamil Nadu

From the 5th year onwards, adopt the following irrigation schedule based on pan evaporation for drip irrigation and basin irrigation.

Western Region of Tamil Nadu

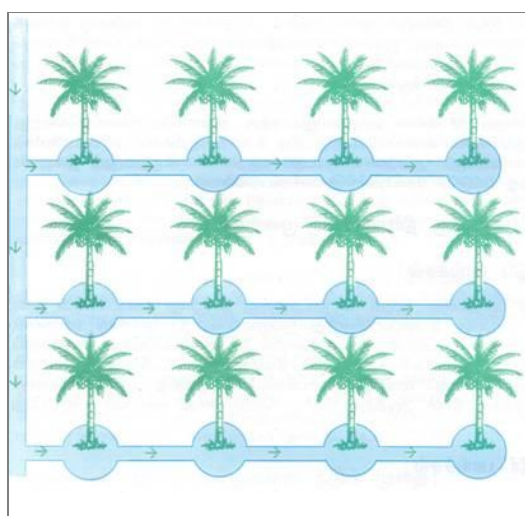
Months	Normal condition (for best yield)	Moderate water scarcity condition	Severe water scarcity condition
A. Drip irrigation			
February to May	65 lit / day	45 lit/ day	22 lit / day
January, August and September	55 lit / day	35 lit / day	18 lit/day
June and July, October to December	45 lit / day	30 lit/ day	15 lit / day
B. Basin irrigation			
February to May	410 lit / 6 days *		
January, August and September	410 lit /7 days*		
June and July, October to December	410 lit /9 days*		

Eastern Region of Tamil Nadu

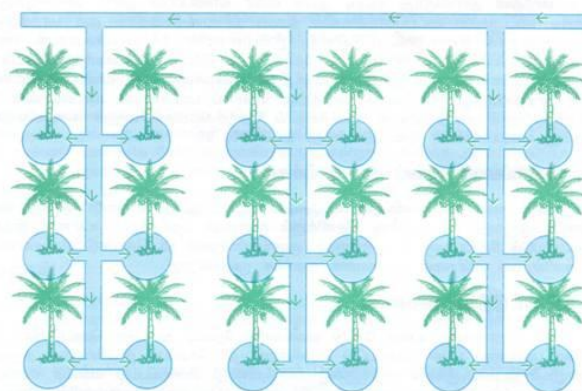
Months	Normal condition (for best yield)	Moderate water scarcity condition	Severe water scarcity condition
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A. Drip irrigation			
March – September	80 lit / day	55 lit / day	27 lit/day
October – February	50 lit / day	35 lit/ day	18 lit /day
B. Basin irrigation			
March – September	410 lit / 5 days*		
October – February	410 lit /8 days*		

* Quantity of water to be applied in the basin. Add 30 - 40 % of the above quantity of water (135 -165 litres/palm) to meet the conveyance loss.



Wrong Method: Irrigation by a Single Channel



Correct Method: Irrigation by Main and Sub Channels

[↑Top of page](#)

Kerala

Irrigate the palms during summer months in basins around palms as shown below:

Irrigation Requirement of Coconut

Parameters	Soil texture			
	Sandy	Sandy loam	Loam	Silty clay
Available Soil Moisture (cm/m)	8	12	17	21
Quantity of water / irrigation / palm in litres in a basin of 1.8 m radius	600	900	1300	1600

Frequency of irrigation (days)				
All areas in Kerala except north eastern portion of Thrissur and Palakkad districts	3-4	5	7-8	9
North eastern portion of Thrissur and Palakkad districts	2-3	3-4	5-6	6-7

Note: In coastal sandy soils, seawater can be used for irrigation. Do not irrigate seedlings and very young palms upto 2 year with sea water. In irrigated gardens, interruption of irrigation would lead to serious setback in yield and general condition of palms. Hence, when once started, irrigation should be continued regularly and systematically. Drip irrigation is the best suited method of irrigation for coconut. It saves water, labour and energy.

[↑Top of page](#)

Karnataka

Irrigation and soil moisture conservation

- Coconut palm responds to summer irrigation. Production of female flowers and setting % increases considerably due to irrigation. Since spadix initiation to ripening of nuts takes nearly 42 months, the full benefit of irrigation can be felt only after 3 years.
- Burying the Coconut husk or coir dust is one of the most effective ways of conserving soil moisture.
- These husks coir dusts can act as sponge and absorb and retain moisture about 6-10 times respectively to their own weight and slowly release it to the coconut trees during dry periods.
- As the husk or dust breaks down slowly, their effect will last for 4-6 years and 8-10 years respectively. On decomposition they also add potash to the soil. These husks or dusts can be added in pits / trenches taken in between the trees but in all the cases depth should be 0.6m and 1.8m away from the bole. Husks / dusts can be added in alternate layers with soil.
- Each palm requires 55 to 120 litres of water every day. Since availability of irrigation water is scanty, for judicious utilization of this resource, adoption of drip irrigation system is most ideal.
- Drip irrigation is known to save about 30 – 40% water with 38% - 40% increase in yield compared to basin irrigation system apart from conserving soil and reducing the competition from weeds for water and nutrients. Through fertigation, we can achieve efficient utilization of both water and nutrients.

Drip Irrigation: In the traditional system of irrigation followed in coconut gardens such as flood irrigation, basin irrigation etc. irrigation efficiency is only 30 to 50 per cent due to considerable wastage of water. In addition, cost on inputs like labour and energy in adopting these systems are high. Scarcity of water and increasing cost of labour and energy are deterrents in adopting these traditional irrigation systems. Under these circumstances, drip irrigation is the most suitable system of irrigation to coconut.

Some of the major advantages of drip irrigation are:

- It saves water
- Enhances plant growth and yield
- Saves energy and labour, most suited for soils having low water holding capacity and undulating terrain
- Reduces weed growth and improves efficiency of fertilizers.

For coconut, generally, three to four drippers are given per palm. For drip irrigation, open four pits of size of 30 x 30 x 30 cm opposite to each other at one meter distance from the trunk. Place 40 cm long PVC pipe (16 mm) in a slanting position in each pit and place the drippers inside the tube and allow the water to drip 30 cm below the soil surface. Fill the pits with coir pith to prevent evaporation. The cost of drip system including installation will be Rs. 130 to 150 per palm (exclusive of pump) which works out to Rs. 23000 to 26000/- approximately per hectare of coconut garden with 4 emitters per palm.

Fertigation: Application of 50% of the recommended dose of fertilizer through drip fertigation produces a yield equivalent to 100% of the recommended dose of fertilizer applied through conventional method. The fertilizers are applied through bypass tank to the palms. Fertilizers namely 70g Urea, 60g DAP and 170g Muriate of potash are recommended for single dose per palm (Like this 6 doses are to be given to the palms which are to be applied from Dec – May at monthly intervals for Kerala condition). For phosphorus application commercial phosphoric acid can also be used.

Drought Management and Soil Moisture Conservation: Coconut produces nuts round the year. Therefore, adequate supply of water is essential for its unhindered growth. Soil moisture is essential for the absorption of nutrients by roots. Moisture stress leads to stunted growth, drooping of leaves, immature nut fall and decreased yield. Mulching is an effective method of conserving soil moisture. Mulch the coconut basins with green / dry leaves at the close of northeast monsoon (October-November). Mulching also adds organic matter to the soil and reduces the soil temperature. Do not disturb soil in the coconut garden during summer months. In level lands, during rainy seasons excess water may be conserved in small trenches dug out in the plantation. In sloppy areas, land may be terraced and trenches dug across. This will facilitate maximum percolation of rainwater and water

storage. For moisture conservation, lowermost 3-5 leaves may be cut and removed. Provide adequate shade for the transplanted seedlings for 1-2 years. To minimize the heat load on the stem, application of lime solution on the trunk up to a height of 2-3 m at the start of the summer season is recommended.

a. Mulching with Coconut Husks/Leaves/Coir Pith: Apply coconut husks with convex surface facing upwards (100 Nos.) or dried coconut leaves (15 Nos.) or coir pith up to a height of 10 cm in the basin of 1.8 m radius around the palms as mulch for soil moisture conservation particularly during summer season.

b. Burial of Coconut Husk or Coir Pith: Husk burial can be done in coconut basins or in the interspaces to overcome drought and button shedding. Bury husks @ 100 Nos. with concave surface facing upwards or 25 kg of coir pith /palm in circular trenches, dug 30 cm width and 60 cm depth at 1.5 – 2.0 metres radius. The husk can be also buried in the trenches at a distance of 3 m from the palm with a size of 45 cm deep and 150 cm width in between two rows of coconut. The soaking of the coconut husk or coir pith as the case may be preserves the monsoon rains. The beneficial effect of husk burial will last for about 5-7 years.